TABLE I
Master Consolidated Benchmark Results: 5-Class KDIGO vs. 3-Class Triage Formulations ($N = 374$, 25 Outer Test Folds)

Target Task	Model Architecture	QWK [95% CI]	Macro-F1 [95% CI]	Accuracy	MAE	Within ±1 [%]
5-Class KDIGO	Logistic Regression (L2)	0.8021 [0.788, 0.815]	0.5419 [0.523, 0.559]	54.2%	0.551	92.6%
	Logistic Regression (Elastic-Net)	0.8029 [0.789, 0.816]	0.5414 [0.524, 0.558]	54.0%	0.551	92.6%
	SVM-RBF (Calibrated)	0.7885 [0.775, 0.801]	0.4434 [0.431, 0.455]	47.2%	0.637	90.6%
	Balanced Random Forest	0.8121 [0.803, 0.822]	0.4963 [0.472, 0.521]	49.9%	0.579	93.3%
	HistGradientBoosting (Balanced)	0.8147 [0.805, 0.826]	0.4985 [0.481, 0.520]	50.0%	0.576	93.4%
	Continuous Log-eGFR Regressor	0.8204 [0.811, 0.830]	0.5236 [0.502, 0.546]	51.0%	0.549	95.2%
	Calibrated Soft-Voting Ensemble	0.8068 [0.797, 0.816]	0.5087 [0.496, 0.522]	51.0%	0.577	92.8%
3-Class Triage	Logistic Regression (L2)	0.8208 [0.803, 0.839]	0.7393 [0.716, 0.764]	77.8%	0.235	98.8%
	Logistic Regression (Elastic-Net)	0.8253 [0.808, 0.843]	0.7413 [0.719, 0.765]	78.0%	0.231	98.9%
	SVM-RBF (Calibrated)	0.8098 [0.791, 0.828]	0.7130 [0.691, 0.735]	77.3%	0.245	98.2%
	Balanced Random Forest	0.8342 [0.821, 0.848]	0.7322 [0.715, 0.750]	78.1%	0.228	99.1%
	HistGradientBoosting (Balanced)	0.8261 [0.817, 0.836]	0.7075 [0.697, 0.719]	77.2%	0.240	98.9%
	Continuous Log-eGFR Regressor	0.8563 [0.843, 0.871]	0.7701 [0.752, 0.790]	81.5%	0.196	98.9%
	Calibrated Soft-Voting Ensemble	0.8389 [0.834, 0.844]	0.7444 [0.740, 0.749]	78.9%	0.220	99.1%